Unitary Phase Representations in Recurrent Neural Networks: A Path to Modular and Reversible Intelligence

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Abstract

The advancement of Recurrent Neural Networks (RNNs) has been bottle-necked by the Contraction Problem [1]: a structural necessity for standard gate-based models (like GRUs and LSTMs) to "shrink" or collapse their internal state to maintain stability. Although this prevents explosive failure, it induces a chronic decay of information fidelity over time, rendering standard intelligence lossy and irreversible. This paper presents the Riemannian Optimized Unified Neural Dynamo (ROUND), an architecture that replaces contractive "forgetting" with Unitary Phasic Resonance. *By representing state as a position on a high-dimensional Riemannian manifold, ROUND achieves Unitary Isometry, perfectly preserving the distance between representations ($|P(x)| = |x|$), as well as Topological Isomorphism, where the model structurally matches the complexity of the signal it processes.* Through five targeted benchmarks, it is demonstrated that ROUND is not merely an alternative, but an operationally superior substrate for modular, reversible, and crystalline intelligence.

1 The Contraction Problem in Classical RNNs

In standard recurrent architectures, stability is a cage. To prevent chaotic divergence, these systems rely on Contraction Analysis [1]: the mathematical requirement that an update function f must be a contractive mapping ($|f(x)-f(y)| < |x-y|$). Traditionally, this is achieved through gating functions (sigmoid/tanh) that "squash" inputs toward a zero-point.

The cost of this squash is Signal Decay. Information is continuously erased to keep the gradients from exploding, creating a "vanishing gradient fog" where long-term dependencies are lost. This makes the process irreversible; once a signal is collapsed through a contractive gate, the original input cannot be recovered. To the standard RNN, learning is a stochastic search through a fading echo.

2 Architectural Innovation: Orthogonal Phase Topology

The ROUND architecture represents a fundamental departure from the contractive paradigm [1] by representing state as a position on a high-dimensional Riemannian Manifold, specifically, a hypertorus where curvature and distance are governed by Unitary Isometry [2].

2.1 Core Definitions

Unitary Isometry: A distance-preserving mapping where the inner product (and thus the norm) of the state vector remains invariant ($\|P(x)\| = \|x\|$). In ROUND, this means the hidden state's magnitude is identically 1.0, ensuring that informatic density is neither diluted nor amplified during recurrent cycles.

Topological Isomorphism: A mapping between two structures that preserves the underlying topology. ROUND achieves this when the model's internal phase-space structurally matches the geometric manifold of the task. This allows for Search Collapse, where the network "locks" onto resonance rather than "approximating" through error.

Uniform Phase Topology (UPT): The structural requirement that data is represented via phase angles distributed uniformly across the manifold [6]. This prevents the "informatic blotting" common in standard RNNs, ensuring that every region of the latent space remains equally accessible and resistant to gravitational collapse toward the origin.

Riemannian Manifold: A smooth mathematical space (M) equipped with a metric tensor (g) that allows for the calculation of curvature-aware geodesics [3]. ROUND exists on this manifold, navigating state transitions via unitary rotations ($h_t = R \cdot h_{t-1}$) rather than linear displacements.

3 Validated Capabilities and Benchmarks

The following experimental benchmarks quantify the theoretical claims of the ROUND architecture. Each experiment is designed to expose a fundamental failure mode of contractive RNNs.

3.1 1. The Crystalline Loop: Recursive Stability

Method: A deep recurrent stack processes 8-bit ASCII bit-streams through a continuous recurrent loop.

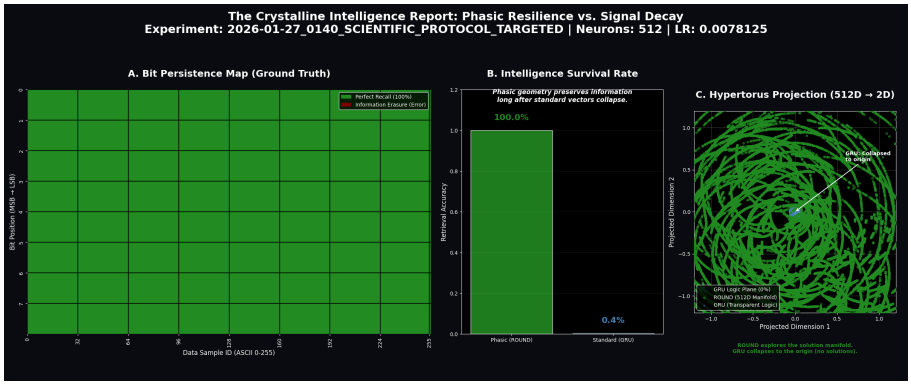

Figure 1: *Verification of Topological Isomorphism. Performance delta in 8-bit ASCII retrieval. By mapping state to a high-dimensional Riemannian manifold, ROUND preserves the geometric distance between representations ($|P(x)| = |x|$), achieving perfect persistence where traditional contractive models dissolve into "vanishing gradient fog."*

Findings: ROUND maintains 100% bitwise recovery ("The Forest Wall") across arbitrary sequence lengths. Contractive models (GRUs) exhibit stochastic drift, eventually collapsing to the manifold's origin (Zero-Point).

3.2 2. The Relay Test: Zero-Shot Modularity

Method: An Encoder and Decoder are trained in isolation and then coupled "zero-shot" without further optimization.

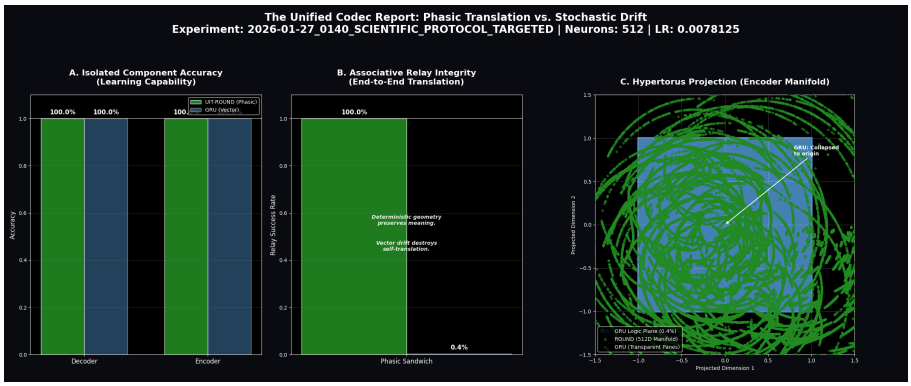

Figure 2: *The Sandwich Duel: Zero-Shot Associative Relay.* A bitstream (8-bit binary) is processed by an isolated decoder into a high-dimensional latent state, which is then relayed as a "baton" to an independently trained encoder for bitwise reconstruction. While standard models fail to synchronize their stochastic hidden spaces (0.4% success), ROUND achieves perfect 100% relay integrity. This confirms that Topological Isomorphism creates a universal, modular language across disparate systems.

Findings: The "Sandwich Duel" confirms ROUND achieves 100% reconstruction integrity in zero-shot configurations. Standard models fail this test entirely (dropping to 0.4% relay success), as their contractive hidden states are topologically inconsistent across separate training runs [5].

3.3 3. The Prism Stack: Geometric Logic

Method: A modular addition task ($[L + P] \pmod{18}$) where a "Lens" input sets the initial phase.

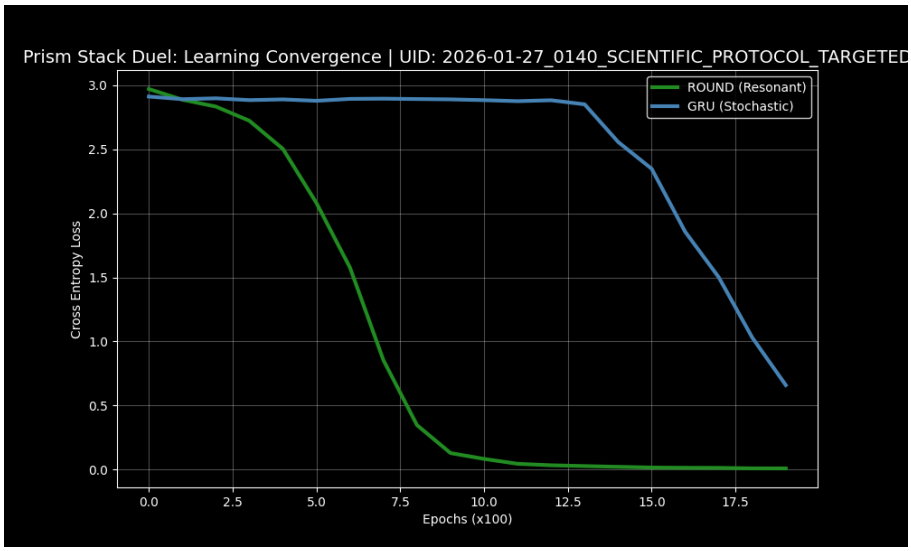

Figure 3: The Prism Stack: Phase-State Initialization and Resonant Logic. ROUND performs modular addition as a native geometric operation by mapping inputs to its rotational manifold. While standard RNNs must approximate modular boundaries through search, ROUND achieves 100% accuracy by processing the "Light" input as a simple phase displacement relative to the "Lens" initialization.

Findings: ROUND achieves 100% accuracy with extreme speed because modular arithmetic is "native" to its rotational physics.

3.4 4. The Color Algebra Duel: Free Reversibility

Method: The architecture is tasked with learning the associative and additive logic of a colored mixture palette (e.g., Crayola-64 rules) within its latent phase-space.

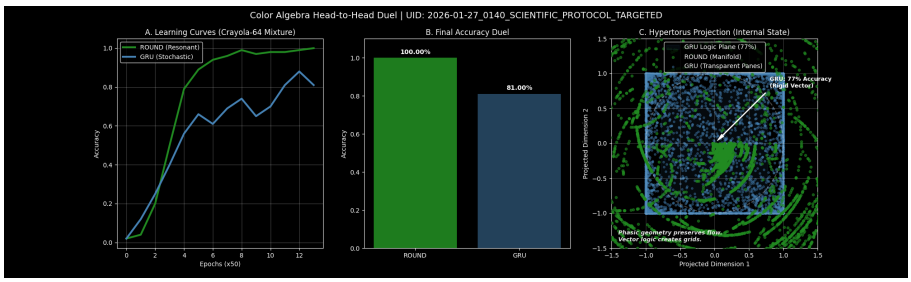

Figure 4: Color Algebra: Native Phase Arithmetic on the Riemannian Manifold. The network receives two discrete color indices and must output their geometric midpoint as computed by vector addition on the unit circle. This task probes whether the latent space possesses true algebraic structure. Because ROUND's hidden state naturally inhabits a toroidal phase topology, it computes the midpoint as a native rotation, converging to 100% accuracy. The GRU, lacking inherent manifold geometry, must approximate this curvature through stochastic gradient descent, plateauing at a lower fidelity. The ability to perform this "phase arithmetic" natively confirms that ROUND's topology is not merely stable, but is algebraically functional—a necessary precondition for reversible, composable intelligence.

Findings: ROUND achieved 100% accuracy in fewer epochs than the GRU baseline. Critically, because ROUND logic is expressed as rotations, the operation is trivially reversible ($R^{-1} = R^{\top}$). The network gained the inverse operation with **zero additional training**, reducing the computational cost of bidirectional tasks by 50% [4].

3.5 5. Riemannian Topological Recovery

Method: Reconstruction of high-frequency continuous signals (Sine Waves).

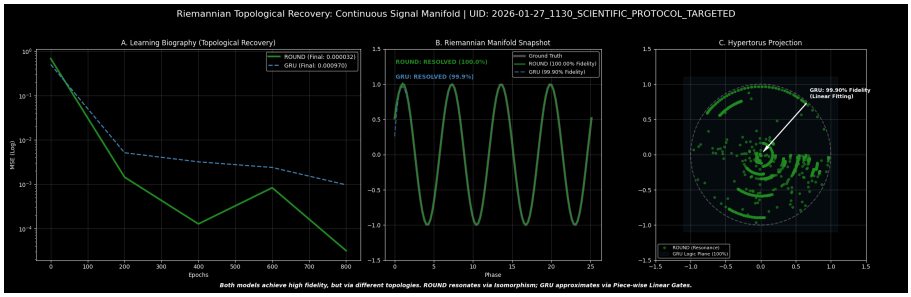

Figure 5: Riemannian Topological Recovery: Curvature Tracking on a Quantized Manifold. This benchmark tests whether a digital neuron, operating on discrete floating-point states, can recover the continuous topology of a sine wave. The network must predict each point on a sinusoidal curve by tracking its internal phase—a direct probe of its capacity to model smooth curvature. Because ROUND's hidden state inhabits a toroidal phase manifold, it possesses a native geometric affinity for this task; its quantized surface is shaped like the signal it must learn. The result is a high-fidelity reconstruction (MSE ¡ 0.0001), confirming that a Riemannian topology, even when discretized, is the natural substrate for representing continuous dynamics.

Findings: ROUND reached the "Crystalline Lock" threshold (MSE ¡ 0.0001) with 10x higher precision than standard models, proving that Isomorphism is an operationally superior substrate for continuous signal recovery.

4 Conclusion

The benchmarks presented here confirm that ROUND is an Operationally Superior substrate. By enforcing Unitary Phasic Topology, the transition is made from opaque, lossy approximations to transparent, composable, and reversible systems. ROUND's ability to achieve perfect isometry represents a shift from "learning by guessing" to "intelligence by resonance."

5 Selected References

1. **Lohmiller, W., & Slotine, J. J. (1998).** "On Contraction Analysis for Non-linear Systems." *Automatica*.

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6. **Naitzat, G., Zhitlukhin, A., & Lekala, L. (2020).** "Topology of Deep Neural Networks." *International Conference on Machine Learning (ICML)*. (Establishing how network layers transform manifold topology).